Purchase Receiving Exception Manager

A Microsoft Dynamics 365 Business Central AL extension · illustrated walkthrough

The business problem

Deliveries don't always match the purchase order: fewer units arrive, goods are damaged, or the wrong item is sent. Standard Business Central records what was received, but it has no structured place for *why* a delivery was wrong, and nothing stops the invoice from being posted while the problem is still being investigated. This extension adds that, on top of the standard purchasing process and without changing it.

Standard Business Central vs. this extension

Standard Business Central 28.5	Added by the extension
Purchase orders, partial receipts (Qty. to Receive / Quantity Received)	Exception record per problem: reason, supplier delivery reference, affected quantity, description
Receipt and invoice posting (codeunit 90 "Purch.-Post")	One narrow check before posting: invoicing an order is blocked while an Open exception requires invoice review. Receiving is never blocked
Nothing links a delivery problem to the order	"Open Exceptions" count on the Purchase Order, with actions to view and create exceptions
Standard permission sets for purchasing	Two roles on top: PRX RECEIVING (create) and PRX MANAGER (resolve through a controlled operation only)
Order is deleted after it is fully posted	Exception keeps its own copy of vendor, item and delivery details, plus who resolved it, when and why
—	Report "Open Purchasing Exceptions by Vendor"

How it was built and tested

Platform	Business Central 28.5 on-premises (US), CRONUS USA demo company, SQL Server Express, Windows Server 2022 VM
Code	AL (runtime 17.0), compiled with CodeCop, UICop and PerTenantExtensionCop: 0 errors, 0 warnings
Automated tests	22 tests in the AL Test Tool, all passing (creation rules, posting restriction, permissions)
Real users	Two Windows users, receiver1 and manager1, each with standard purchasing permission sets plus one extension role. Every demo screenshot on pages 2–3 was taken signed in as one of them, not as an admin

The demo on the next two pages

Order 20 handbags → record the problem → receive the 12 that arrived → invoice blocked → receiving user cannot resolve → manager resolves → invoice posts → history kept

Scenario (synthetic data): a retailer orders 20 tan tote handbags from *Riyadh Leather Supplies*. Only 12 arrive, but the supplier's delivery note DN-4471 says the shipment was complete. Purchase order 106028, exception PEX-00005, posting date 1/27/2028 (CRONUS work date).

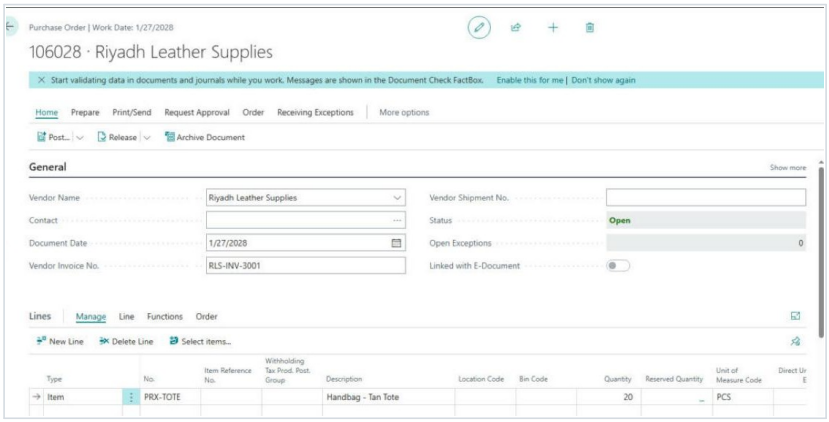
The demo, part 1: the receiving clerk (receiver1)

1

Purchase order for 20 handbags.

signed in as receiver1

A standard BC purchase order. The only visible additions are the **Open Exceptions** field (0) and the **Receiving Exceptions** tab.

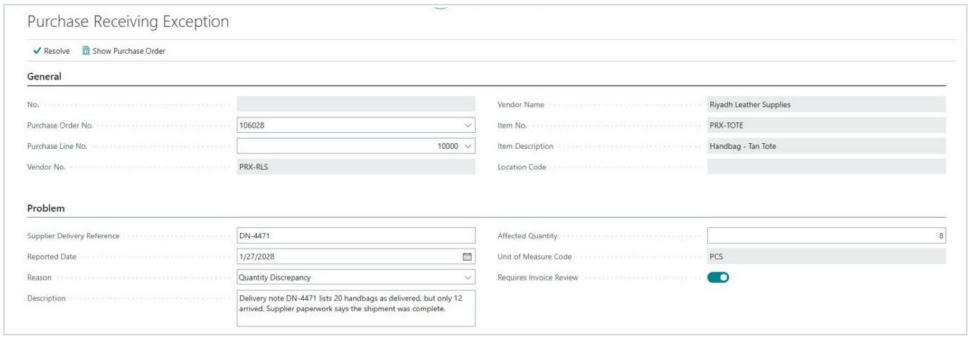


2

The receiving clerk records the problem.

signed in as receiver1

New Exception from the order pre-fills the order and vendor; choosing line 10000 fills item and unit of measure from the order. The clerk enters reason, delivery reference DN-4471, affected quantity 8 and turns on **Requires Invoice Review**.

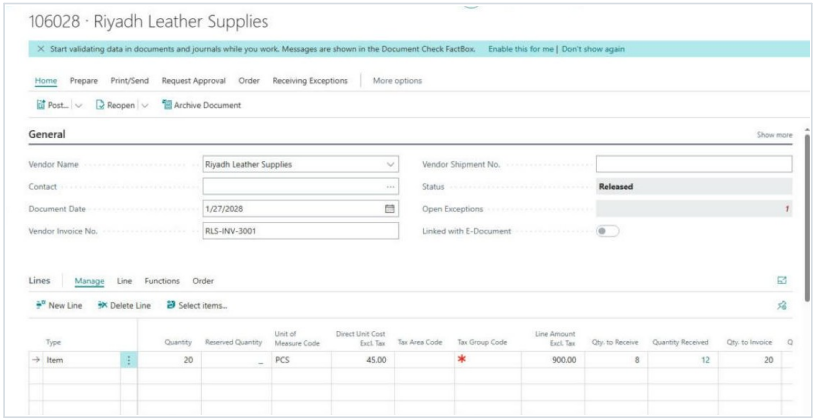


3

The 12 accepted units are received with standard posting.

signed in as receiver1

Receiving is never blocked. The line shows Quantity 20, Quantity Received 12, Qty. to Receive 8, so 8 are still outstanding. Open Exceptions = 1.



4

Invoice posting is blocked.

signed in as receiver1

Posting the invoice for the 12 received stops before anything is posted. BC shows the extension's message on its Error Messages page (full text below): it names the order, the blocking exception and what to do next.

"Purchase order 106028 cannot be invoiced yet. Receiving exception PEX-00005 (Quantity Discrepancy: Delivery note DN-4471 lists 20 handbags as delivered, but only 12 arrived. Supplier paperwork says the shipment was complete.) is Open and requires invoice review. Ask a purchasing manager to review and resolve it, or post the receipt only."

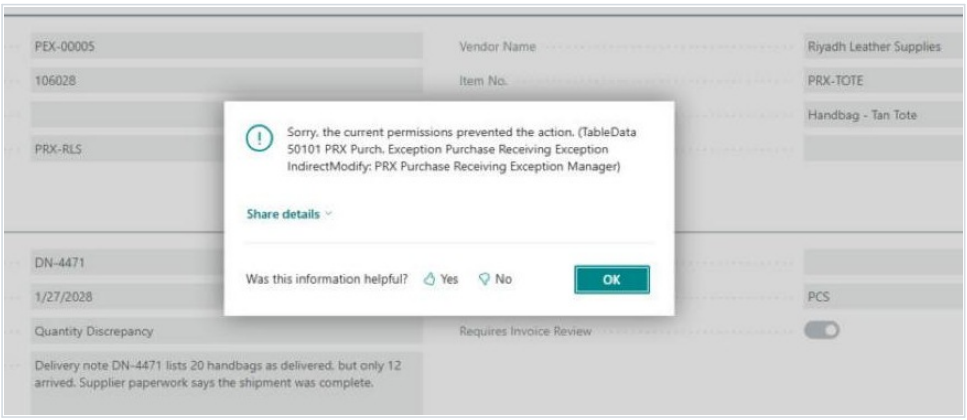
The demo, part 2: permissions, resolution and posting

5

The receiving user cannot resolve (permission proof).

signed in as receiver1

receiver1 can open the Resolve dialog, but the server refuses the change: the role has no modify right on the exception table, so hiding a button is not what protects it.

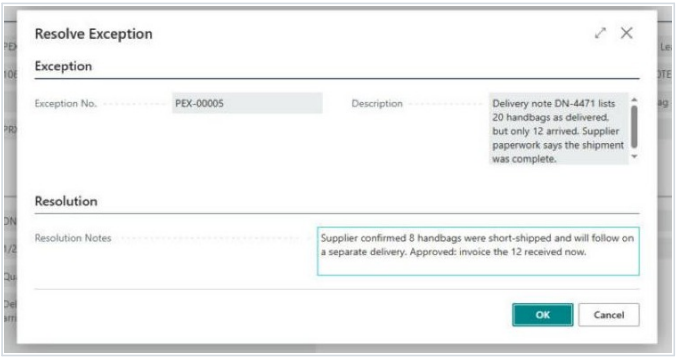


6

The purchasing manager resolves it, with an explanation.

signed in as manager1

Resolving requires notes (an empty note is rejected). The change can only happen inside the extension's Resolve operation, which records who and when.

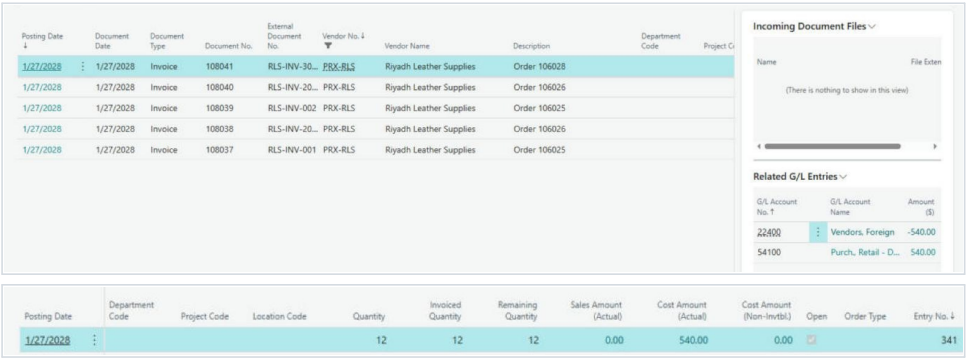


7

The invoice posts, and the ledger entries are standard BC.

signed in as manager1

Posted invoice 108041 for order 106028: vendor ledger entry with G/L 540.00 (12 × 45.00); item ledger entry 341: Quantity 12, Invoiced 12, Cost Amount (Actual) 540.00. The extension writes no ledger entries itself.



8

The exception history stays on record.

signed in as manager1

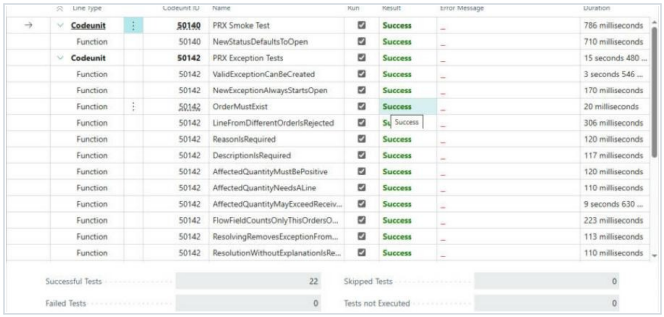
Created By RECEIVER1 at 1:28 AM, Resolved By MANAGER1 at 1:44 AM, with the resolution notes, the original description and the delivery reference.

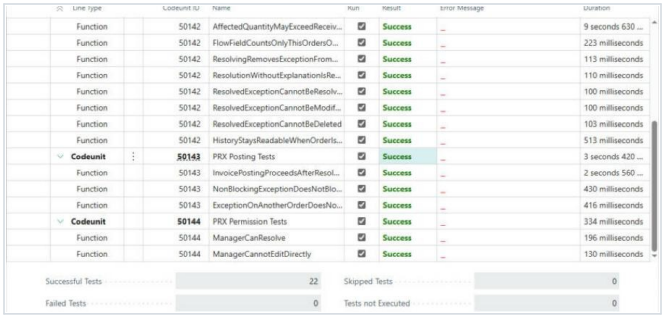


Evidence and limitations

Automated tests: 22 of 22 passing

AL Test Tool (page 130451) on the deployed build. Covered: creation rules (order must exist, line must belong to the order, reason and description required, affected quantity rules), the open-exception count, resolution rules, resolved records read-only, invoice posting after resolution, exceptions on other orders not blocking, and permissions (Permissions Mock).





What is in the extension

Tables	Exception (with number series via the "No. Series" module) and Setup
Table extension	Purchase Header: "Open Exception Count" FlowField
Pages / page extension	Exception list, card and Resolve dialog, setup page; Purchase Order extension (count + actions)
Codeunits	Management (validation), Resolve (controlled change), Install, Demo data
Event subscriber	Purch.-Post, OnBeforePostPurchaseDoc: the invoice-review check; standard posting is unchanged
Permission sets	PRX RECEIVING and PRX MANAGER (indirect modify, usable only through the Resolve codeunit)
Report	Open Purchasing Exceptions by Vendor (RDLC layout)
Tests	Separate test app: helper library + 4 test codeunits

Known limitations

- The block is order-level: one review-required exception also stops invoicing other lines on the same order.
- Only purchase-order posting through "Purch.-Post" was tested; separate purchase invoices, warehouse receipts and other posting routes were not verified.
- No links to posted receipts or invoices, and no reopen or cancel: an exception goes Open → Resolved only.

Source code and documentation

github.com/HaseebAhmed2003/Purchase_receiving_exception

Source for the extension and the test app, installation steps, test overview and known limitations. I can also run the demo live on my laptop and make changes on request.